Microeconomics I

Class 5 — Indirect utility, wealth and price effects

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13 October 2026

Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Short quiz
10:10	Warm-up; rice in Hunan
10:20	Indirect utility
10:41	Roy's identity
10:50	Break
11:00	Wealth effects: normal and inferior goods
11:14	Price effects: Giffen goods, substitutes, complements
11:29	In pairs: from v to demand; summary

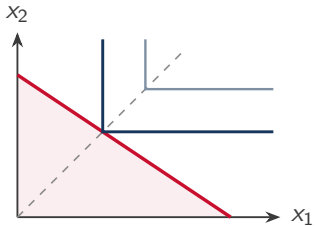
Practicalities

- Problem set 1: due at the start of class tomorrow, Wed 14 Oct, on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445

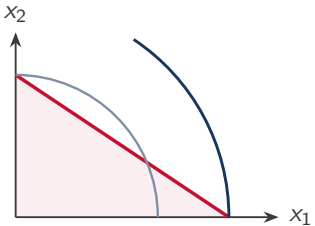
  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: where does each consumer buy?

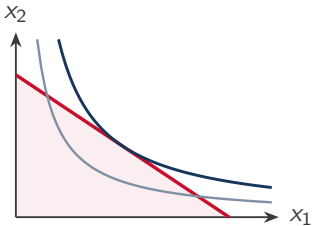
$p = (2, 3)$, $w = 12$ in every panel



L-shaped, kinks on $x_2 = x_1$



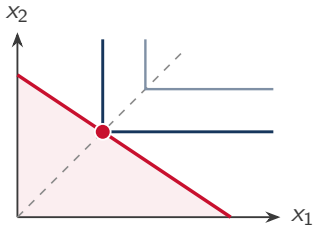
bowed out



$x_1 x_2$ **or** $\ln x_1 + \ln x_2$

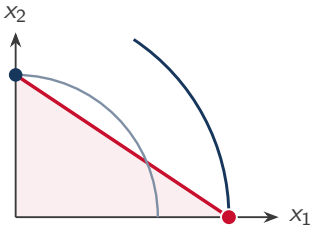
Warm-up: where does each consumer buy?

$p = (2, 3)$, $w = 12$ in every panel



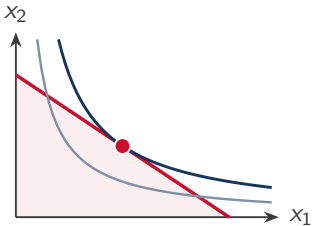
L-shaped, kinks on $x_2 = x_1$

the kink, not a tangency: $(2.4, 2.4)$



bowed out

a corner, and it jumps: $(0, 4)$ once $p_1 > 3$



$x_1 x_2$ **or** $\ln x_1 + \ln x_2$

same bundle, another number: 6 or $\ln 6$

Kinks and corners need no tangency; the formula changes the number, not the bundle.

Rice gets cheaper. Do poor households eat more rice?

Hunan, China, 2006: 650 poor urban households

rice: 64 % of their calories

at random, vouchers: up to a quarter
off the price of rice, for five months

More rice, the same, or less?

Jensen & Miller (2008), American Economic Review 98(4): 1553–1577.

Cheaper rice, less rice — for the poor, not the poorest

poor, but not the poorest

at least 20 % of calories from other food

$+0.45$

% more rice for a 1 % higher price

the poorest

over 80 % of calories from rice

< 0

cheaper rice, more rice

For the poor but not the poorest, demand for rice slopes up: why, after the break.

Jensen & Miller (2008), "Giffen Behavior and Subsistence Consumption", AER 98(4): 1553–1577, Table 3.

Indirect utility

MWG 3.D, 3.G

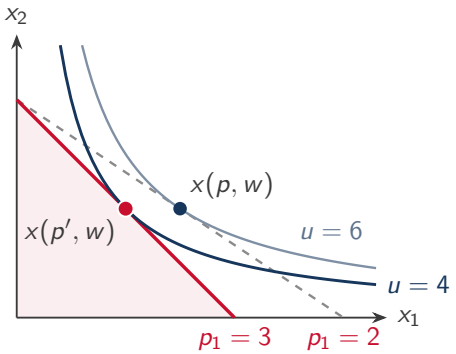
Indirect utility: the level of the best curve the budget reaches

$v(p, w) = u(x^*)$ for any $x^* \in x(p, w)$

the value of the UMP: $\max\{u(x) : x \in \mathcal{B}_{p,w}\}$

$u = x_1 x_2$: $\quad v(p, w) = \dfrac{w}{2p_1} \cdot \dfrac{w}{2p_2} = \dfrac{w^2}{4p_1 p_2}$

at $p = (2, 3)$: $w^2/24$ (class 4, slide 26)



p_1: $2 \to 3$: v falls from 6 to 4.

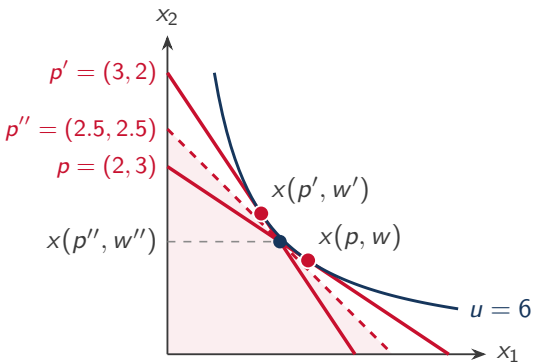
MWG 3.D, p. 56.

$v(p, w)$ is homogeneous, rises with w and never rises with prices

property	why	for $v = \dfrac{w^2}{4p_1p_2}$
(i) homogeneous of degree 0	the same budget set	$\dfrac{(\alpha w)^2}{4\,\alpha p_1\,\alpha p_2} = v$
(ii) strictly increasing in w	LNS (class 2)	$\dfrac{\partial v}{\partial w} = \dfrac{w}{2p_1p_2} > 0$
nonincreasing in each p_ℓ	the budget set shrinks	$\dfrac{\partial v}{\partial p_1} = -\dfrac{w^2}{4p_1^2p_2} < 0$
(iii) quasiconvex in (p, w)	next slide	
(iv) continuous		

MWG Proposition 3.D.3, p. 56: u continuous, $\succsim$ locally non-satiated, $X = \mathbb{R}^2_+$.

v is quasiconvex: the average of two budgets never beats both



$\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for every $\bar{v}$

$p = (2, 3)$ or $p' = (3, 2)$, $w = 12$: $v = 6$ both

their average $(2.5, 2.5)$, $w = 12$: $v = 5.76$

What the average budget affords, one of the two already affords: $v \leq 6$.

MWG Proposition 3.D.3(iii), p. 56; Figure 3.D.5, p. 57, in Wednesday's reading; $u = x_1 x_2$.

In pairs: could this be an indirect utility function?

Test (i) and (ii) of Prop. 3.D.3 (slide 9). Four minutes.

$v(p,w) =$	homogeneous of degree 0	$\uparrow$ in w, $\downarrow$ in p
$\frac{w}{p_1 + p_2}$		
$\frac{w^2}{p_1 p_2}$		
$\frac{p_1 + p_2}{w}$		
$\ln w - \ln p_1 - \ln p_2$		

In pairs: could this be an indirect utility function?

Test (i) and (ii) of Prop. 3.D.3 (slide 9). Four minutes.

$v(p, w) =$	homogeneous of degree 0	$\uparrow$ in w, $\downarrow$ in p	
$\frac{w}{p_1 + p_2}$	✓	✓	$u = \min\{x_1, x_2\}$ (class 4)
$\frac{w^2}{p_1 p_2}$	✓	✓	four times slide 8's v
$\frac{p_1 + p_2}{w}$	✓	×	more wealth, less utility
$\ln w - \ln p_1 - \ln p_2$	×	✓	changes with the unit of money

Rows 3 and 4 fail: no consumer has them.

Change the utility function: v and λ change, demand does not

u	$v(p, w)$	$\lambda = \partial v / \partial w$ at $p = (2, 3)$, $w = 12$
$x_1 x_2$	$\dfrac{w^2}{4 p_1 p_2}$	$\dfrac{w}{2 p_1 p_2} = 1$
$\frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$	$\ln w - \frac{1}{2} \ln p_1 - \frac{1}{2} \ln p_2 - \ln 2$	$\dfrac{1}{w} = \dfrac{1}{12}$

Same $\succsim$, same $x(p, w) = (3, 2)$; v becomes $f(v)$, and λ with it.

MWG p. 57, in Wednesday's reading ($\tilde{u} = f(u)$ gives $\tilde{v} = f(v)$; Example 3.D.2 with $\alpha = \frac{1}{2}$); $\lambda = \partial v / \partial w$: pp. 54–55, class 4, slide 26.

Roy's identity: demand from two slopes of v

$$x_\ell(p, w) = -\frac{\partial v(p, w)/\partial p_\ell}{\partial v(p, w)/\partial w}$$

u continuous; $\succsim$ locally non-satiated and strictly convex; v differentiable at $(p, w) \gg 0$

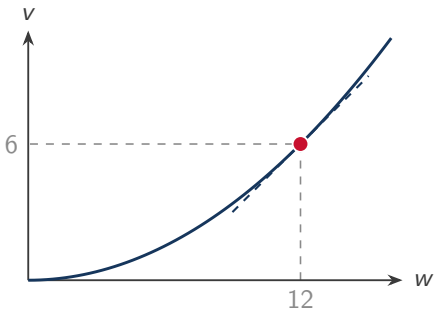
$$v = \frac{w^2}{4p_1p_2}\colon \quad -\frac{-w^2/(4p_1^2p_2)}{w/(2p_1p_2)} = \frac{w}{2p_1}$$

Cobb–Douglas, class 4

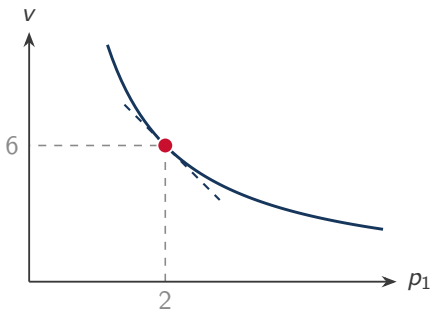
Replace u by $f(u)$: both slopes gain the factor f', and the ratio does not change.

MWG Proposition 3.G.4 (Roy's identity), p. 73.

A dearer good costs utility in proportion to how much of it she buys



$p = (2, 3)$: slope $\lambda = 1$



$w = 12$, $p_2 = 3$: slope $-\lambda x_1 = -3$

$$\frac{\partial v}{\partial w} = \lambda, \qquad \frac{\partial v}{\partial p_\ell} = -\lambda\, x_\ell(p, w)$$

Here $x_1 = -(-3)/1 = 3$: the bundle $(3, 2)$ again.

MWG p. 74, Proofs 2 and 3 (the envelope theorem); $u = x_1 x_2$, $p = (2, 3)$, $w = 12$.

(optional) A bliss point: does more wealth always raise v?

$u(x) = -(x_1 - 2)^2 - (x_2 - 2)^2$ on $\mathbb{R}^2_+$, $p = (2, 3)$.

(1) $v(p, w)$ for $w \geq 10$.

(2) Which part of Prop. 3.D.3 fails?
Why?

(3) $\lambda = \partial v / \partial w$ at $w = 12$?

MWG Proposition 3.D.3, p. 56 (slide 9).

(optional) A bliss point: does more wealth always raise v?

$u(x) = -(x_1 - 2)^2 - (x_2 - 2)^2$ on $\mathbb{R}^2_+$, $p = (2, 3)$.

(1) $v(p, w)$ for $w \geq 10$.

$v = 0$: $(2, 2)$ costs 10

(2) Which part of Prop. 3.D.3 fails? Why?

(ii): $v(p, 10) = v(p, 12)$;
LNS (Def. 3.B.3) fails at $(2, 2)$

(3) $\lambda = \partial v / \partial w$ at $w = 12$?

$\lambda = 0$: the budget does not bind

MWG Proposition 3.D.3, p. 56 (slide 9).

Break

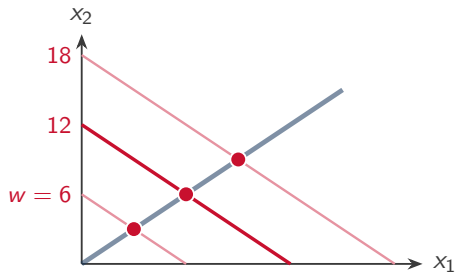
Back at 11:00

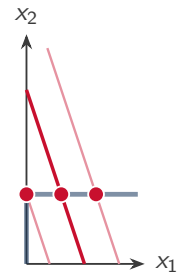
Wealth and price effects

MWG 2.E

More wealth: Cobb–Douglas scales up; quasilinear: all to good 1

Engel function: $w \mapsto x(p, w)$, p fixed; wealth effect $\partial x_\ell / \partial w$





$u = x_1 x_2$, $p = (2, 3)$

$x = (w/4,\ w/6)$: a ray

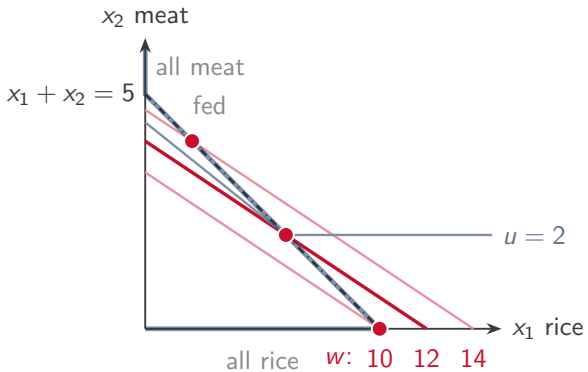
$u = x_1 + \ln(1 + x_2)$, $p = (3, 1)$

$x_2 = 2$ once $w \geq 2$

Class 4's promise: past $w = 2$, every extra euro goes to good 1.

MWG 2.E, pp. 24–25, Figure 2.E.1 (the wealth expansion path).

Rice can be inferior: past subsistence, more wealth buys less of it



$p = (2, 3)$

she needs 5 units of food:
$x_1 + x_2 \geq 5$

once fed, only meat counts

normal at (p, w): $\partial x_\ell / \partial w \geq 0$;
inferior: < 0

Rice is normal up to $w = 10$, inferior from 10 to 15.

$u = \min\{x_2,\ x_2 - 4(5 - x_1 - x_2)\}$: meat, minus 4 per unit of missing food. MWG 2.E, p. 25; own example.

The budget ties the effects together: not every good can be inferior

$$\sum_k p_k \frac{\partial x_\ell}{\partial p_k} + w \frac{\partial x_\ell}{\partial w} = 0 \qquad \text{homogeneity: Prop. 2.E.1}$$

$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial w} = 1 \qquad \text{Walras in } w\text{: Engel aggregation, Prop. 2.E.3}$$

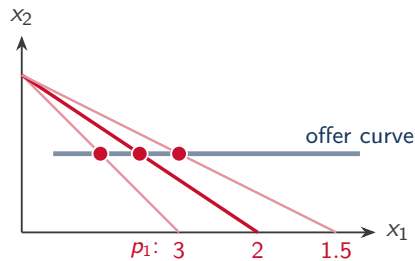
$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial p_k} + x_k = 0 \qquad \text{Walras in } p_k\text{: Cournot aggregation, Prop. 2.E.2}$$

Slide 19 at $w = 12$: $2 \cdot (-1) + 3 \cdot (+1) = 1$. An extra euro is spent somewhere.

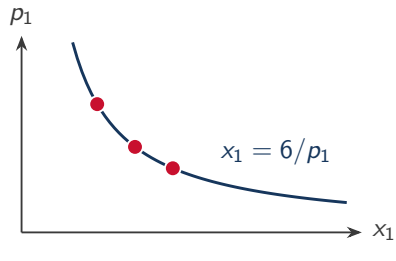
MWG 2.E, pp. 27–28.

Cheaper good 1: Cobb–Douglas buys more of it, as much of good 2

usual: $\partial x_\ell / \partial p_\ell \leq 0$; Giffen at (p, w): $\partial x_\ell / \partial p_\ell > 0$



goods: the offer curve

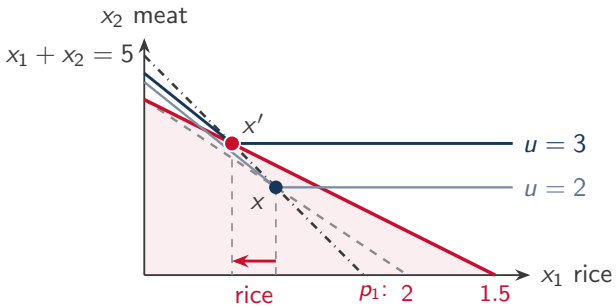


prices: the demand curve

$x_1 = 6/p_1$ rises as p_1 falls; x_2 stays at 2: the offer curve is flat.

MWG 2.E, pp. 25–26, Figures 2.E.2–2.E.3; $u = x_1 x_2$, $w = 12$, $p_2 = 3$.

Cheaper rice, less rice: the subsistence consumer is Giffen



p_1: $2 \to 1.5$

rice $3 \to 2$, meat $2 \to 3$

on the food line: $x_1 = 3/(3 - p_1)$ rises with p_1

Giffen needs inferior (slide 19)

Cheaper rice frees money for meat, which also feeds her.

Jensen & Miller (2008), AER 98(4): their subsistence model; own numbers, $w = 12$, $p_2 = 3$.

Dearer good 2: x_1 falls, stays or rises, depending on σ

gross substitutes: $\partial x_1/\partial p_2 > 0$; gross complements: < 0

σ	u	$x_1(p, w)$	p_2: $3 \to 4$ $p_1 = 2$, $w = 12$
0	$\min\{x_1, x_2\}$	$\dfrac{w}{p_1 + p_2}$	$2.4 \to 2$: complements
1	$x_1 x_2$	$\dfrac{w}{2p_1}$	$3 \to 3$: neither
2	CES, $\rho = \frac{1}{2}$	$\dfrac{w\, p_2}{p_1(p_1 + p_2)}$	$3.6 \to 4$: substitutes

CES: complements for $\sigma < 1$, neither at $\sigma = 1$, substitutes for $\sigma > 1$.

MWG 3.G, p. 70 ("gross" = Walrasian demand; the Hicksian version is Wednesday). CES as class 4, slide 29.

In pairs: Roy's identity, then the three effects

Find the demand by Roy's identity. Wealth effect? Own price: usual or Giffen? The other price rises: substitutes or complements? Seven minutes.

$v(p, w)$	demand	wealth	own price	other price ↑
$\frac{w^3}{p_1 p_2^2}$, find x_1				
$\frac{w + p_2}{p_1} - 1 + \ln \frac{p_1}{p_2}$, find x_2 class 4's $x_1 + \ln(1 + x_2)$, $p_1 > p_2$				
$\frac{w - 5p_1}{p_2 - p_1}$, find x_1 slide 22's consumer, near $p = (2, 3)$, $w = 12$				

In pairs: Roy's identity, then the three effects

Find the demand by Roy's identity. Wealth effect? Own price: usual or Giffen? The other price rises: substitutes or complements? Seven minutes.

$v(p, w)$	demand	wealth	own price	other price ↑
$\frac{w^3}{p_1 p_2^2}$, find x_1	$x_1 = w/(3p_1)$	normal	usual	neither
$\frac{w + p_2}{p_1} - 1 + \ln \frac{p_1}{p_2}$, find x_2 class 4's $x_1 + \ln(1 + x_2)$, $p_1 > p_2$	$x_2 = p_1/p_2 - 1$	zero: normal in MWG	usual	substitutes
$\frac{w - 5p_1}{p_2 - p_1}$, find x_1 slide 22's consumer, near $p = (2, 3)$, $w = 12$	$x_1 = \frac{5p_2 - w}{p_2 - p_1}$	inferior	Giffen	substitutes

Row 3 is slide 22's rice: inferior and Giffen, from one quotient.

(optional) Good 1 is Giffen: what happens to x_2 when p_1 rises?

A consumer with two goods; Walras' law holds; $\partial x_1/\partial p_1 > 0$.

(1) Show that $\partial x_2/\partial p_1 < 0$.

Hint: slide 20, line 3, with $k = 1$.

(2) Explain (1) with her spending.

(3) Can $\partial x_1/\partial p_2 > 0$ hold too?

MWG 2.E, pp. 27–28 (Proposition 2.E.2, Cournot aggregation).

(optional) Good 1 is Giffen: what happens to x_2 when p_1 rises?

A consumer with two goods; Walras' law holds; $\partial x_1/\partial p_1 > 0$.

(1) Show that $\partial x_2/\partial p_1 < 0$.

Hint: slide 20, line 3, with $k = 1$.

$p_2\,\partial x_2/\partial p_1 = -x_1 - p_1\,\partial x_1/\partial p_1 < 0$

(2) Explain (1) with her spending.

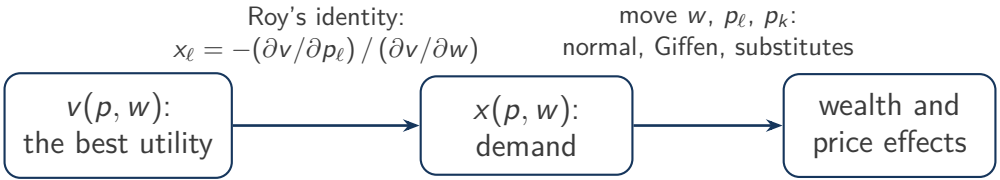
p_1 and x_1 rise, so $p_1 x_1$ rises and $p_2 x_2$ falls

(3) Can $\partial x_1/\partial p_2 > 0$ hold too?

Yes: slide 22's rice at $p_1 = 2$:
$\partial x_1/\partial p_2 = 2$, $\partial x_2/\partial p_1 = -3$

MWG 2.E, pp. 27–28 (Proposition 2.E.2, Cournot aggregation).

Today in one line



Wednesday: expenditure minimisation — Hicksian demand, Shephard's lemma, duality.

Problem set 1: due at the start of class tomorrow, on paper, one copy per group.

Problem set 2 is handed out tomorrow; due in class Wednesday 21 October.

Read MWG 3.E–3.G (or Muñoz-Garcia ch. 2) for Wednesday 14 October.